

# Simultaneous Linear Equations - Nathaniel Juan Putra W

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## Student

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Topic: Simultaneous Linear Equations

Status: completed

## Lesson Plan

### Learning Objectives

- Solve simultaneous linear equations using the methods of substitution and elimination.
- Represent simultaneous linear equations graphically to identify points of intersection.
- Construct a system of linear equations from complex word problems to model real-world scenarios.
- Evaluate the efficiency of different solving methods based on the structure of given equations.

### Lesson Information

### Lesson Sections

#### 1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Nathaniel! Great to see you. Before we dive into today's challenge, let's keep those logarithm skills sharp. I've put four quick checks on the board. For the first two, we're looking at the Change of Base formula. For the last two, use your product and quotient rules. Go ahead and type the final simplified values in the chat as you get them!

Activities:

- Check 1: Evaluate  $\log_2(50)$  using base 10 (approximate).
- Check 2: Simplify  $(\log 9) / (\log 3)$  without a calculator.
- Check 3: Expand  $\log(x^2 y)$  using properties.
- Check 4: Condense  $2 \log(x) - \log(y)$  into a single term.

## 2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Imagine you're managing a tech startup. You have two different pricing plans for your software, and you need to find the exact point where the costs for your customers align, or where your supply perfectly meets demand. This 'point of balance' is what we find using simultaneous equations. Today, we're moving from single-variable puzzles to dual-variable systems. Ready to find the intersection?

Activities:

- *Discussion on the 'Intersection' concept using a Business Supply/Demand graph.*

## 3. Review of Prior Knowledge (5 minutes)

Teacher Script:

To solve these, we need to be experts at isolating variables. If I give you  $2x + y = 10$ , how quickly can you write  $y$  in terms of  $x$ ? Excellent. We will use this 'substitution mindset' as our first primary weapon today.

Activities:

- *Quick-fire variable isolation: Solve  $3x - 2y = 12$  for  $y$ .*

## 4. Problem Introduction (10 minutes)

Teacher Script:

Let's look at a 'Hard' difficulty system:  $3x + 4y = 18$  and  $5x - 2y = 4$ . We can't just see the answer. We have two main paths: Substitution (making one variable the 'subject') or Elimination (making the coefficients of one variable match and adding/subtracting). Which looks more efficient here? Since we have a  $+4y$  and a  $-2y$ , multiplying the second equation by 2 to eliminate ' $y$ ' seems like a pro move. Let's try that together.

Activities:

- *Demonstrating the Elimination Method step-by-step.*
- *Visualizing the two lines on a coordinate plane.*

## 5. Guided Practice (15 minutes)

Teacher Script:

Now, let's tackle a word problem. 'A tech store sells 2 types of VR headsets. 3 units of Type A and 2 units of Type B cost \$1200. 5 units of Type A and 4 units of Type B cost \$2100.' First, Nathaniel, let's define our variables. What should  $x$  and  $y$  be? Now, help me build the two equations. Walk me through the elimination steps.

Activities:

- *Translating word problems into  $2 \times 2$  systems.*
- *Collaborative solving on the whiteboard.*

## 6. Independent Practice (15 minutes)

Teacher Script:

You're flying solo now! I've placed three challenges on the board. The first is a straightforward substitution, the second requires elimination with two multipliers, and the third is a tricky word problem involving speeds. I'll be watching your work live—take your time with the signs!

Activities:

- Solving: 1)  $y = 2x - 3$  &  $5x - 3y = 11$ ; 2)  $7x + 2y = 47$  &  $5x - 4y = 1$ ; 3) Modeling a real-world interest rate problem.

## 7. Consolidation & Reflection (5 minutes)

Teacher Script:

Amazing work today. You solved systems that even some adults find intimidating! Remember, the 'solution' is just the coordinate where two stories meet. Before you go, don't forget to log in to your dashboard at <https://learn.algonova.id/login> to access your review materials and today's homework. What was the 'aha!' moment for you today?

Activities:

- Self-reflection on the choice of method (Substitution vs Elimination).

## Homework

### Medium Questions:

1. Solve using substitution:  $x + y = 15$ ;  $y = 2x$ .
2. Solve using elimination:  $2x + 3y = 13$ ;  $2x - y = 1$ .
3. A shop sells notebooks for \$3 and pens for \$2. If 10 items were bought for \$26, how many of each?
4. Find the intersection of  $y = 4x - 5$  and  $y = -x + 10$ .

### Hard Questions:

1. Solve:  $3x + 4y = 25$ ;  $2x + 3y = 19$ .
2. Solve:  $0.5x - 0.2y = 1.3$ ;  $0.3x + 0.4y = 2.3$ .
3. Find the values of  $a$  and  $b$  if the line  $ax + by = 10$  passes through  $(2, 1)$  and  $(4, -2)$ .
4. A boat goes 30km upstream in 3 hours and 30km downstream in 2 hours. Find the speed of the boat in still water and the speed of the current.
5. Solve:  $(2/x) + (3/y) = 13$ ;  $(5/x) - (4/y) = -2$  (Hint: use substitution  $u=1/x$ ,  $v=1/y$ ).
6. Determine if the system  $2x + 3y = 7$  and  $4x + 6y = 14$  has one, none, or infinite solutions. Justify your answer.

### Answer Key:

Medium 1:  $(5, 10)$ . Medium 2:  $(2, 3)$ . Medium 3: 6 notebooks, 4 pens. Medium 4:  $(3, 7)$ .

`Hard 1: (x=1, y=5.5). Hard 2: (x=3.77, y=2.92 approximately or exact fractions). Hard 3: a=3, b=4. Hard 4: Boat=12.5 km/h, Current=2.5 km/h. Hard 5: x=1/2, y=1/3. Hard 6: Infinite (lines are identical/coincident).`

`REVIEW: Change of Base -  $\log_2(10) = \log(10)/\log(2)$  "H 3.32.`

`REVIEW: Change of Base -  $(\log 100)/(\log 10) = 2$ .`

`REVIEW: Properties -  $\log(x/y^2) = \log x - 2\log y$ .`

`REVIEW: Properties -  $3\log 2 + \log 5 = \log(8*5) = \log 40$ .`

## Parent Reminder

### Today's Math Progress: Simultaneous Equations

Nathaniel tackled complex simultaneous linear equations today! He focused on the elimination and substitution methods and practiced modeling real-world business scenarios with algebra. He also reviewed high-level logarithm properties to ensure long-term retention. Great focus today!